Case Study: Factor Analysis of Mental Ability Test Scores

Undergraduate Statistics Program | Factor Analysis in R & SPSS

Background

- Psychologists and educational measurement specialists often use multiple tests to measure broader latent abilities that cannot be observed directly. For example, several spatial tasks may reflect a visual-spatial ability factor, several language tasks may reflect a verbal/textual factor, and several timed tasks may reflect a processing-speed factor. Because the observed test scores are correlated, factor analysis can be used to identify a smaller number of underlying dimensions that summarise the pattern of student performance across tests.
- In this case study, you are working as a statistician supporting an educational assessment team. Your task is to use exploratory factor analysis to determine how many latent ability dimensions are supported by the test scores, interpret those factors, and assess whether the final model adequately reproduces the observed correlation matrix.
- The Holzinger–Swineford 1939 dataset is a classic dataset in psychometrics and factor analysis. It contains mental ability test scores for students from two schools, Pasteur and Grant-White. In the version commonly distributed with the R package lavaan, the dataset contains 301 observations and 15 variables. For this assignment, you will analyse the nine continuous test-score variables x1 to x9.

Dataset

Use the following variables for the factor analysis. Treat them as approximately continuous numerical test scores. The expected domains above are provided only to help with interpretation after the analysis. Students should still justify the retained number of factors using statistical evidence rather than assuming the answer in advance.

Variable	Test description	Expected domain to consider
x1	Visual perception	Visual / spatial ability
x2	Cubes	Visual / spatial ability
x3	Lozenges	Visual / spatial ability
x4	Paragraph comprehension	Textual / verbal ability
x5	Sentence completion	Textual / verbal ability
x6	Word meaning	Textual / verbal ability
x7	Speeded addition	Processing speed
x8	Speeded counting of dots	Processing speed
x9	Speeded discrimination of straight and curved capitals	Processing speed

Assignment objective

Your task is to analyse the Holzinger–Swineford mental ability test-score data in R or SPSS and produce a short statistical report. You should use factor analysis to study the relationships among the nine test-score variables, identify and interpret the retained latent factors, compare alternative rotations, evaluate model fit, and use factor scores to compare groups of students.

By the end of the assignment, students should be able to:

- compute and comment on descriptive statistics and correlations;
- assess whether the data are suitable for Factor Analysis using KMO and Bartlett's test;
- perform Factor Analysis in R or SPSS;
- interpret factor loadings, communalities, uniquenesses, explained variance and factor scores;
- compare different rotations;
- translate statistical results into a substantive interpretation of Greek municipalities.

Task 1. Prepare and describe the dataset.

Load the Holzinger–Swineford 1939 dataset and create a working dataset containing the nine test-score variables x1 to x9. Briefly describe the source of the data, the unit of analysis, and why the nine variables are appropriate for factor analysis.

Present useful descriptive statistics for the nine variables, including mean, standard deviation, minimum, maximum, skewness, and kurtosis. Include histograms or boxplots for selected variables. Comment on whether the variables look approximately continuous and whether there are unusual outliers or strong departures from normality.

Task 2. Examine the correlation structure of the variables.

Compute and present a correlation matrix for the selected variables. You may also use scatterplot matrices or other suitable plots. Use formal diagnostic tools to assess whether the dataset is suitable for Factor Analysis. Compute the Kaiser-Meyer-Olkin measure of sampling adequacy and Bartlett's test of sphericity.

Task 3. Decide how many principal components should be retained.

Investigate how many latent factors should be retained. Use and compare multiple criteria. Justify your final choice.

Task 4. Fit and interpret the Factor Analysis model.

Fit the final Factor Analysis model using maximum likelihood estimation. Report and interpret:

- factor loadings;
- communalities;
- uniquenesses;
- variance explained by each factor;
- the test of whether the chosen number of factors is sufficient.

Task 5. Compare rotations and name the latent factors.

Fit at least two rotated solutions, for example varimax. Compare the loading patterns and discuss which rotation gives the clearest substantive interpretation. Because cognitive abilities may be correlated, an oblique rotation is often plausible. If you use an oblique rotation, report the factor correlation matrix and comment on it.

Use the rotated factor loadings to identify groups of variables that appear to measure similar latent dimensions. Give substantive names to the retained factors, such as visual-spatial ability, textual/verbal ability, or processing speed, if supported by your results.

Task 6. How good is the factor model?

Evaluate how well the final factor model reproduces the observed correlation matrix. Compute the residual correlation matrix.

Task 7. How can municipalities be compared using factor scores?

Compute factor scores from the final model and add them to the original dataset. Use the scores to compare groups of students rather than focusing on named individuals. For example, compare mean factor scores by school, grade, or gender, depending on the variables available in your working file.

Create at least one table or plot showing how groups differ on the retained factors. Interpret the comparison carefully: factor scores are estimated quantities and should be used descriptively unless a formal inferential comparison is also performed.

Required deliverables

No.	Deliverable	Description
1	Short report (5–8 pages)	Business context, descriptive analysis, FA results, and interpretation.
2	R script or R Markdown / Quarto file or SPSS .spv output file.	Fully reproducible code, clearly commented, with clean data preparation and labelled outputs.
3	Executive summary (7-8 ppt slides)	A concise representation of the analysis and the results.

Assessment criteria

- Correctness of the statistical analysis.
- Quality of interpretation and business reasoning.
- Clarity of tables, plots, and written discussion.
- Reproducibility and readability of the R/SPSS code.

Interview relevance

- This case mirrors the kind of work often discussed in consulting, analytics, and strategy interviews: handling correlated indicators, simplifying a complex decision problem, and translating technical outputs into an expansion recommendation for management.